Exact reductions and quantitative bounds for mutually unbiased bases in dimension six

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Abstract

This technical checkpoint separates finite exact calculations, reported Lean implications, and unresolved links to concrete mutually unbiased bases. A pinned replay of the finite-corner classification, together with concrete normalization and transport, now reduces actual quartet existence to completion of a retained reconstruction. A separate exact Cauchy–Binet calculation gives a minor with squared determinant at least $54/5$ and inverse-entry squared moduli at most $10/27$ in every raw Hadamard. Whether nonextension holds for every raw Hadamard remains open. The bounded minor now also supplies a complete block reconstruction chart. Clearing its nonzero determinant denominator yields explicit division-free matrix equations equivalent to actual quartet existence, with both additional bases retained. A checked affine completion of the balanced transition matrices gives an equivalent system of 648 rational polynomial equations of degree at most two in 320 bounded real coordinates. Its infeasibility is not established. The code also contains cyclotomic fixtures, rational interval routines, conditional atlas and gluing interfaces, and finite parity calculations. The Route C obstruction still needs a certified Fourier-triple landing premise. The declared 56-cell cover still needs analytic curvature certificates. The operator interface admits a zero-action countermodel and therefore needs stronger mathematical definitions before it can express the intended exclusion theorem. Separate concrete constructions now supply the continuous two-circulant family, variable-basis equations, and an actual fourth-basis candidate from normalized positive flat compatible moments. The needed empty varieties and full analytic cover remain open. No unconditional dimension-wide MUB bound follows from this package at this checkpoint.

1. Mathematical problem and normalization

This article presents reductions and quantitative bounds, rather than a solution of the dimension-six MUB problem. Its contribution is a separation of concrete matrix reductions, finite calculations, conditional interfaces and remaining exclusion obligations. Section 2.1 compares the reductions with existing family results and the finite-corner classification.

Formal-verification statements below describe the source reports identified in Section 6. The standalone preprint archive contains the manuscript and rendering machinery; it contains no Lean project or computational research supplement. Those formal statements have not received a complete fresh replay for this release candidate and are not certified by its checksum verifier.

Let B_a be a unitary 6×6 matrix whose columns form an orthonormal basis of $\mathbb{C}^6$. Mutual unbiasedness means

$$|(B_a^* B_b)_{ij}|^2 = \frac{1}{6} \qquad (a \neq b).$$

The unnormalized transition $H_{ab} = \sqrt{6} B_a^* B_b$ then has unit-modulus entries and $H_{ab}^* H_{ab} = 6I$. All computational checks must retain this normalization; testing an unnormalized product against the normalized condition changes the problem. A complete family would have seven bases and $\binom{7}{2} = 21$ unordered transitions. A fourth-basis obstruction and a seven-basis obstruction are distinct statements.

MubQuartetTransportBridge.lean now derives this normalization for every actual quartet. exists_quartet_iff_exist s_rawQuartet connects the existing normalized matrix predicate to three raw Hadamard matrices with all three MU conditions. MubMonomialTransport.lean derives preservation under a common left row-permutation/phase action and independent right column actions. Thus standard Hadamard equivalence preserves the entire completion problem. no _quartet_of_concrete_representatives still requires both complete representative coverage and nonextension for every representative. The separate pinned MubFiniteCornerReduction.lean now consumes the verified external coverage

theorem and derives an equivalence between actual quartet existence and raw quartet completion of some retained finite-corner output. Its nonexistence equivalence leaves the exclusion of every retained output as the remaining mathematical statement, rather than assuming coverage.

The Lean type MUB6.Gluing.Basis and its MU relation are opaque interfaces. Their names do not supply coordinates, unitary matrices, or the above equations. A theorem over those interfaces needs a separate concrete interpretation before it can be applied to the Hilbert-space problem.

1.1. Gauge coordinates for a fixed first Hadamard matrix

Apply the same left unitary B_0^* to every basis. Then $B_0 = I$ and $H_a = \sqrt{6}B_a$, for $a = 1, 2, 3$, have unit entries in modulus and $H_a^* H_a = 6I$. A common left diagonal phase action, followed by independent column phases, dephases H_1. The common left action must be applied to all bases; independently changing rows of just one basis need not preserve MU.

For a fixed H_1, right multiplication of H_2, H_3 by diagonal matrices with entries $\overline{(H_a)_{0j}}$ fixes their first rows to one. Entry moduli, unitarity and all transition moduli are preserved. The acting column-phase group is $\mathbb{T}^{12}$, of real dimension **12**. The prescription fixes 24 ambient real coordinates: twelve phases and twelve already-constrained radial coordinates. Thus $144 - 24 = 120$ real coordinates remain for the two additional raw matrices. This is an ambient coordinate count, not a dimension theorem for the solution set, a smoothness assertion, or a Jacobian rank claim. There are still sixty circle equations before imposing orthogonality and MU. If H_1 also varies, its dephased 5×5 block contributes another fifty ambient real coordinates; the fixed-H_1 count must not be used globally.

The matrix-level preservation statements are SzollosiColumnGauge.columnRephase_gram, columnRephase_unitary, columnRephase_rawPairMU and rawCandidates_firstRowDephase. The family-specific existence equivalence is exists_gauged_solution_with_parameters_iff; it keeps the family parameters fixed and both additional bases variable.

Scope. This supports the stated gauge reduction. It does not classify H_1, establish a 120-dimensional manifold, or exclude a quartet. The narrow claim is preservation of the completion problem under column phases. The next required evidence is a complete obstruction for the remaining equations over a declared domain, including the parameters when they vary.

1.2. Explicit transition-balance identities

Write $A = H_2$, $B = H_3$, $W = A^* B$ and define polynomial residuals

$$g_{Aj} = A_{0j} - 1, \quad g_{Bj} = B_{0j} - 1, \qquad R_{Bm} = \sum_k B_{mk}\overline{B_{0k}} - 6\delta_{m0}.$$

For arbitrary complex matrices A, B, without assuming any residual vanishes, the following identity holds:

$$\sum_k W_{jk} - 6 = \sum_m \overline{A_{mj}} R_{Bm} - \sum_{m,k} \overline{A_{mj}} B_{mk} \overline{g_{Bk}} + 6\overline{g_{Aj}}. \tag{1}$$

Indeed, substituting $\overline{B_{0k}} = 1 + \overline{g_{Bk}}$ cancels the double sums on the right, leaving $\sum_{m,k} \overline{A_{mj}} B_{mk} - 6\overline{A_{0j}} + 6(\overline{A_{0j}} - 1)$. Interchanging A, B and conjugating gives the column identity. On the first-row gauge and row-unitarity equations both residuals vanish, so all six row sums and six column sums of W are six. These are twelve complex, or twenty-four real, equations.

Equation (1), split into real and imaginary parts, is an explicit polynomial ideal-membership identity for the row-unitarity and gauge residuals. Its coefficients depend on matrix entries. For a square matrix, column unitarity also implies row unitarity, as formalized by SzollosiGaugeBalance.raw_right_gram. The solution-level conclusions are transition_row_sum, transition_column_sum and solution_balance_equations. No additional solution-set restriction follows by adding these consequences, although they can strengthen an interval contractor on boxes containing nonsolutions.

Scope. MubGaugeSyzygy.lean now checks the row identity and its conjugate column counterpart over every commutative star ring. Its two ideal-membership corollaries apply to any ideal containing the displayed generators, including polynomial ideals. They support exact algebraic redundancy, without supplying a Jacobian rank or nonextension result. All six displayed theorem endpoints were replayed on 2026-10-11 with only standard Lean axioms. The next step is to bind these generator identities to the executable contractor's coordinates.

1.3. A quantitative invertible minor

For any $A \in \mathbb{C}^{3\times 6}$ with $AA^* = 6I_3$, the finite Cauchy–Binet identity gives

$$\sum_{S\subset\{0,\dots,5\},\ |S|=3} |\det A_{[:,S]}|^2 = \det(AA^*) = 216.$$

There are twenty summands. Hence one selected minor $B = A_{[:,S]}$ satisfies $|\det B|^2 \geq 216/20 = 54/5$, in particular $\det B \neq 0$. If all entries of A have unit modulus, each entry of $\operatorname{adj} B$ is a difference of two unit-modulus products. Its squared modulus is at most four, and therefore

$$|(B^{-1})_{ij}|^2 \leq \frac{4}{|\det B|^2} \leq \frac{10}{27}.$$

MubMinorChart.lean checks the twenty-minor identity as an exact polynomial identity and derives the two rational bounds. It also applies them to the first three rows of every raw Hadamard matrix satisfying $HH^* = 6I_6$. All ten displayed endpoints passed the canonical Lean 4.33.1 replay with only standard axioms. The next subsection supplies placement and complete reconstruction coverage. Neither result establishes finite fibres or excludes any quartet. The selected well-conditioned minor need not be the minor chosen by the external finite-corner theorem.

1.4. A complete bounded chart and division-free quartet equations

MubBoundedBlockChart.lean extends each selected triple to a complete column permutation, then dephases the entire matrix while preserving determinant moduli and quartet extendability. On the block index set $\mathrm{Fin}(3) + \mathrm{Fin}(3)$ the resulting matrix is

$$H = \begin{pmatrix} E & B \\ C & D \end{pmatrix}, \qquad q = |\det B|^2 \geq 54/5, \qquad D = -CE^*(B^{-1})^*.$$

AdmissibleEBC requires unit entries in all four blocks, first row and column one in E, first row one in B, first column one in C, and the two diagonal row-Gram equations $EE^* + BB^* = 6I_3$ and $CC^* + DD^* = 6I_3$. The forced formula makes the mixed Gram blocks zero. The module checks soundness, coverage and actual quartet transport. It does not use the external classification or a finite-fibre premise.

MubBoundedQuartetEquations.lean defines

$$d = \overline{\det B}, \qquad N = -CE^*(\operatorname{adj} B)^*, \qquad S = \begin{pmatrix} dE & dB \\ dC & N \end{pmatrix}.$$

It shows in this model $dD = N$ and $dH = S$ under the positive determinant bound. The forced-block constraints become $|N_{ij}|^2 = q$ and $qCC^* + NN^* = 6qI_3$, without division. Retain the top Gram equation, the unit entries of E, B, C, their gauge constants and $q \geq 54/5$. For two further raw matrices U, V, retain unit entries and $U^*U = V^*V = 6I_6$. Independent column phases fix their first rows to one. The three MU conditions are exactly

$$|(S^*U)_{ij}|^2 = 6q, \qquad |(S^*V)_{ij}|^2 = 6q, \qquad |(U^*V)_{ij}|^2 = 6.$$

The checked endpoint exists_quartet_iff_equations identifies actual quartet existence with a solution of these equations and the determinant inequality. The matching nonexistence equivalence is checked too. There are 76 free complex

entries after the gauge constants are substituted, or 152 ambient real coordinates. Unit entries bound each real and imaginary part by one; this is not a solution-dimension or independence claim. The scalar bound and all 34 displayed endpoints across these two modules passed targeted canonical replay with only standard axioms.

MubBoundedCoordinates.lean checks both directions of the 152-real-coordinate codec and retains the full box $[-1,1]^{152}$, including its boundary. MubBoundedPolynomials.lean represents complex arithmetic in the quadratic algebra over $\mathbb{Q}[x_1,\dots,x_{152}]$ with generator squared equal to -1. Its real and imaginary components give 396 indexed rational polynomial equalities. Together with $q \geq 54/5$, their real solutions are equivalent to actual quartet existence (exists_quartet_iff_polynomials). The 13 coordinate and 26 polynomial endpoints passed scoped replay with only standard axioms.

MubBoundedExpressions.lean identifies each executable arithmetic expression with its polynomial, including determinant and adjugate evaluation. The polynomials have total degree at most 8, and q has degree at most 4. MubBoundedCoefficientExpansion.lean identifies finite rational term-list expansion with those same polynomials. These degree and expansion statements passed scoped kernel replay; they do not assert sharp degree or independence.

MubBoundedExportIndices.lean checks complete, unique coordinate and equation numbering. MubBoundedTerm Collection.lean checks polynomial preservation under sorting, coefficient collection and exact-zero removal. MubB oundedCodedSystem.exists_quartet_iff_coded retains actual quartet existence on Fin152 with all 396 equations and the determinant inequality. The 28 displayed endpoints passed scoped replay with only standard axioms. The native export contains 3,760,338 collected monomials across the 396 constraints and q; all 1,588 exact rational expression/list comparisons pass. The export receipt binds the compressed artifact and its decompressed bytes.

MubBoundedMUFactor.lean checks a further cancellation: 36 MU residuals factor as q times polynomials of degree at most four. Since $q \geq 54/5$, replacing those residuals preserves the complete quartet existence equivalence. Their coefficient lists shrink from 371,520 to 3,096 monomials; with every other row retained, the total becomes 3,391,914. The 36-row replacement table is bound to the original archive by its receipt. Eighteen endpoints passed scoped replay. A checked singular-point counterexample shows why the positive determinant condition is required for this cancellation.

MubQuadraticLift.lean supplies a separate full-matrix formulation. Keeping the nine complex entries of D and three complex transition matrices gives 386 ambient real coordinates. MubQuadraticLiftPolynomials.lean checks 648 rational polynomial equalities: 108 unit-entry equations, 216 Gram components, 216 transition graph components and 108 transition norm equations. The graph equations are $W_0 = H^*U$, $W_1 = H^*V$, $W_2 = U^*V$; each transition entry has squared norm six. Actual quartet existence is equivalent to a common zero in $[-1,1]^{170} \times [-3,3]^{216}$, with coordinates grouped by their matrix-entry and transition slots. MubQuadraticLiftDegree.lean bounds the total degree of every actual polynomial by two. The three modules' 28 displayed endpoints passed scoped replay with only standard axioms.

This full-matrix model uses no determinant division. Its reverse implication checks the three full Hadamard matrices and all pairwise MU equations directly, so it does not need invertibility of B. The positive determinant condition remains necessary in the earlier reconstruction and factor-cancellation models. The new coordinate and equation counts are not independence or dimension claims.

MubQuadraticLiftIndices.lean checks complete coordinate and equation numbering and the bounds assigned to coordinate IDs. The coefficient and coded-system modules identify the complete collected lists with these same polynomials and preserve quartet existence on Fin386. The native export contains 4,920 monomials, at most 12 per row, in 169,800 bytes. All 2,592 comparisons with direct rational matrix arithmetic pass, and two exports are byte-identical. Forty-two zero rows remain in the complete indexing.

MubQuadraticLiftTable.lean contains literal rows generated from that file. Its rows_match checks every literal row against the defining coefficient lists by kernel reduction; bounds_match checks the literal bounds. exists_quartet_iff_t able preserves actual quartet existence for common zeros of these literal rows in the coded box. JSON parsing and file hashing remain external steps recorded in the coefficient receipt. This does not identify an arbitrary external file with the checked literals.

The native Rust loader now binds that exact file and checks its schema, indexing, monomials and recorded probe values. One exact interval sweep retains the first 170 coordinate bounds and narrows all 216 transition bounds to $[-10035/4096, 10035/4096]$. Every final equation range contains zero; the result is CONTRACTED_UNRESOL

VED_BOX. Three targeted Rust tests check this result, invalid-input rejection, square enclosure, known-root retention and closed boundary contact. These are native execution checks.

Independently, MubQuadraticLiftNarrowBox.lean derives the same bound from the transition norm equations. Its exists_quartet_iff_narrow_table retains actual quartet existence on the smaller box. All seven displayed endpoints passed scoped replay with only standard axioms. This checks the specific mathematical narrowing without treating the runtime parser or general contractor as kernel-reflected proofs.

MubQuadraticLiftCompatibility.lean derives both transition Gram matrices as $36I$, the three cycle identities $W_0W_2 = 6W_1$, $W_0^*W_1 = 6W_2$, $W_1W_2^* = 6W_0$, and row and column sums equal to six. The polynomial companion adds 504 consequences: 216 column-Gram components, 216 cycle components and 72 linear balance components. Their degrees are checked, and adding them preserves actual quartet existence. These new polynomials have not yet been exported as another literal table.

The separation module checks, for every pair of columns,

$$\sum_k |(W_0)_{ki} - (W_1)_{kj}|^2 = 72 - 12\,\mathrm{Re}(W_2)_{ij} \geq \frac{43623}{1024}.$$

The native inspector uses exact component interval bounds to exclude a box when its squared-distance upper bound is strictly smaller. A closed example with all coordinate intervals of positive width is excluded by this test while the original 648 polynomial ranges still contain zero. In each of two selected columns, real components lie in $[9/10, 11/10]$ and imaginary components lie within $1/20$ of common alternating centers $9/4, -9/4$. The distance upper bound before contraction is $3/10$. close_columns_box_excluded checks this family of regions directly in Lean. Across the three new modules, 37 endpoints passed scoped replay with only standard axioms. Five targeted native tests passed, including the new regional regression and strict-threshold contact. This supplies a regional exclusion; the full box remains unresolved and no complete cover is claimed.

Scope. Infeasibility over the retained domain and a complete exclusion cover remain open. The contraction and coefficient checks do not establish emptiness or solver complexity; $q \geq 54/5$ remains part of the older 152-coordinate system. The earlier large serialized archive is separate from this newly reflected literal table.

The transition balances also permit a global affine reduction. MubBalancedCompletion.lean reconstructs a transition from its leading 5×5 block A: the last column is $6-\sum_j A_{ij}$, the last row is $6-\sum_i A_{ij}$, and the corner is $\sum_{i,j} A_{ij}-24$. Both reconstruction directions are checked without division. MubBalancedQuadraticLift.lean uses these formulas for all three transitions, reducing 386 ambient real coordinates to 320 while preserving actual quartet existence. The retained box is $[-1,1]^{170} \times [-10035/4096, 10035/4096]^{150}$.

The polynomial companion substitutes these affine formulas into all 648 original equations and checks that every resulting degree is at most two. exists_quartet_iff_balanced_polynomials is the resulting global existence equivalence. The recovered coordinate bounds hold on solutions; an arbitrary reduced-box point need not expand into the old box. A checked counterexample uses the all-zero free assignment, whose completed transition corner is -24. All 27 displayed endpoints across the three modules passed scoped replay with standard axioms. Fewer coordinates do not establish a speedup or constraint independence.

The balanced coordinate codec and coefficient substitution now have a complete exact export. The 18 displayed endpoints in MubBalancedIndices.lean, MubBalancedCoefficients.lean and MubBalancedCodedSystem.lean passed scoped replay with standard axioms, including exists_quartet_iff_coded. The generated MubBalancedTable.lean contains literal equality checks in blocks of 18 equations and a proposed exists_quartet_iff_table theorem. Its full build has not completed. Its six displayed endpoints remain unverified; this checkpoint does not claim a checked literal table. The exporter checks 2,592 rational values against direct completed-matrix arithmetic, and two full exports have identical bytes. JSON parsing and byte custody remain external to the kernel.

The balanced table contains 8,313 monomials, compared with 4,920 in the 386-coordinate table. Its largest row contains 676 terms rather than 12; transition-norm equations account for most of the increase. The first 324 rows are unchanged. A corrupted corner row is rejected by the direct matrix check, and the kernel separately checks that its true coefficient list is nonempty. These are implementation and coefficient checks, not exclusion certificates. The existing native inspector

still uses 386 coordinates. Factored completion should be compared with expanded substitution before choosing a reduced search representation; no runtime or contraction-strength advantage has been established. Global infeasibility remains open.

An independent Rust transcription in omega_mub regenerates all 648 rows of the balanced table from the same definitions and compares them with the exported table term by term over the rationals; every row agrees. This is a check of the export against its definitions, not a replay of exists_quartet_iff_table. Two negative controls set the third basis equal to the first or to the second at the product triple. In each case exactly the 36 transition-norm rows of the repeated pair are nonzero, with values -6 and 30, and five core coordinates leave the box. The Jacobian of the system at the centre of the box has 1,045 linear terms in 269 rows and exact rank 184 of 320. It is neither zero nor of full column rank, so a containment test whose premise is a vanishing Jacobian at the centre does not apply to this table, and a Moore–Penrose preconditioner is not defined there. The receipt is artifacts/mub6_prompt05_campaign_receipt.json.

2. Exact arithmetic and finite evidence

The computational fixtures use exact rational or cyclotomic arithmetic over $\mathbb{Q}(\zeta_{24})$. This coefficient field supports those fixtures; it does not restrict arbitrary complex phases in the original problem to 24th roots. Rational interval bounds can support continuum statements only when they enclose the actual matrix-dependent functions on a complete declared domain.

Conjecture2GeneralMechanization.lean contains the finite identities

$$\alpha_2(m) = 6m - 9 \in \{-9, -3, 3, 9\}, \qquad m \in \{0, 1, 2, 3\},$$

$$1 + \text{paritySign}(\alpha_2(m)) = 0,$$

and paired integer cancellations. Its Tao phase-table calculation gives the Eisenstein norm 9. The new balanced_mask_count and tao_all_balanced_masks check all twenty three-positive/three-negative sign patterns among the 64 six-bit masks. For each column the signed exponent is reduced modulo three, and counts (n_0, n_1, n_2) give $x = n_0 - n_2$, $y = n_1 - n_2$, with exact norm $x^2 - xy + y^2 = 9$. tao_distinct_positive_rows also checks every triple of distinct positive row indices directly, without a bit-mask decoding premise. These four theorems were replayed using Lean 4.33.1 on 2026-10-11; #printaxioms reported no axioms for them. MubTaoLaurent.lean now supplies the complex interpretation with $\omega = -1/2 + i\sqrt{3}/2$. It checks addition, multiplication, conjugation and norm under the integer-pair evaluator, identifies inverse entries with their conjugates, and derives

$$\left| \sum_{j=0}^{5} \prod_{r=0}^{5} S_{rj}^{\gamma_r} \right|^2 = 9$$

for every balanced bit-mask sign vector γ. The eight displayed endpoints passed fresh replay using only standard Lean axioms; the integer pair/exponent norm comparison uses none. This is a complex matrix evaluation, not a necessary condition for arbitrary triples. The original module's ModuliClass table is a declared finite table, rather than a classification of complex matrices. Its analytic-certificate structure supplies row-count bounds and table compatibility as assumptions. The later odd-parity polynomial lemmas remain integer identities until a matrix monomial interpretation has been constructed.

The SymPy 2-circulant calculations and Rust non-tensor surveys are separate family evidence. They do not make every complex Hadamard matrix a member of a surveyed family, or establish general Hadamard-cube Conjecture 2.

Scope of the Tao calculation. It supports the exact nonzero value for the recorded matrix and all balanced sign patterns. It does not turn the conjectured Laurent vanishing condition for arbitrary MUB triples into a theorem. The narrow claim is a finite exact invariant evaluation. A Laurent-based exclusion requires that general necessary-condition theorem, separately from the evaluation.

2.1. Literature scope and classification obligations

The Fourier family is continuous, so it should not be grouped as a sporadic matrix. Jaming et al., arXiv:0902.0882v2 [1] establish nonextension to a quartet for the identity together with any member of their Fourier family. Applying that result to an arbitrary transition requires common-unitary and monomial equivalence transport; the new concrete transport modules supply preservation, while matching the exact Fourier representatives to the cited theorem remains a separate obligation.

Maxwell and Brierley, arXiv:1402.4070v2 [2] treat the three-parameter Karlsson family and establish its balanced Laurent vanishing identity. This does not give quartet nonextension. The inherited description of the full Karlsson family as a two-parameter affine family is not used here. Nor is an unspecified matrix named H_6 enough to identify a separate Haagerup exclusion theorem: its concrete entries and source theorem must first be matched.

Cárdenes Wuttig and Tindall, arXiv:2608.18053v2 [3] present a finite-corner classification and an accompanying Lean formalization. Their result concerns single Hadamard matrices; it is not the five-family partition asserted in the inherited campaign. The exact theorem and its equivalence conventions were inspected against the concrete matrix predicates. The source was pinned at 57b03025ecbe259e474281f49e201ea7cb474e34; its 136 project modules passed the source-level unfinished-proof/added-axiom scan. After an initial memory failure and a successful single-worker retry, the public theorem target completed under its pinned Lean 4.33.0-rc2 and Mathlib revision. All nine public theorem reports and the supporting Karlsson coverage report use only standard axioms. All 215 external archive files matched the pinned ZIP.

The integration replay then recompiled the exact source bytes of 21 local dependency modules and the new bridge under the external pin. It reconciles Fin3+Fin3 with Fin6, row and column Gram conventions, and actual monomial equivalence. All five bridge endpoint reports use only standard axioms. The checked conclusion is

$$\begin{aligned}&\exists \text{ an actual quartet in } \mathbb{C}^6\\ &\quad\Longleftrightarrow \exists p, B, C : \ \mathrm{Retained}(p, B, C) \wedge \mathrm{HasRawQuartet}(\mathrm{completion}(p, B, C)).\end{aligned}$$

Here p consists of four unit phases in a dephased corner E; the adjacent blocks obey $EE^* + BB^* = 6I$ and $E^*E + C^*C = 6I$, and the forced final block is $D = -CE^*(B^{-1})^*$. Retention includes the external finite-corner and unit-entry conditions. Finite candidate fibres over continuous seeds do not give a finite global list. Exclusion over every retained reconstruction is still open. The old five-family premise is quarantined.

Matolcsi et al., Proposition 4.2 [4] provide a separate conditional route through their Conjectures 2 and 3. Conjecture 2 concerns Laurent conditions on actual MUB triples; Conjecture 3 concerns quartet nonextension of their Szöllősi family. The paper explicitly withdraws an earlier treatment of the latter as a literature theorem. Thus the local finite class table cannot discharge either premise.

Scope. These source comparisons distinguish established family results, replayed classification, concrete transport, and still-needed exclusion arguments. They do not establish $N(6) = 3$. The narrow conclusion is that the inherited five-branch case split cannot serve as the current proof boundary. The next test is uniform exclusion on the now-justified retained-output domain, or a direct proof of the two conjectures in the alternative route. The independent quantitative minor theorem does not identify its selected minor with the external finite-fibre minor.

3. Gluing statements and the Route C frontier

The combinatorial theorem g3_theorem extracts a fourth basis for any constituent triple of a supplied seven-basis family. The assembly theorem combines supplied landing and obstruction data to contradict that extension. This logical assembly does not construct the landing data.

MUB6OrbitConfiguration assumes

$$6 = n_D + n_T + n_S + n_H, \qquad n_T + n_S \le 1, \qquad n_H \le 0.$$

These natural-number assumptions yield $n_D \ge 5$ and $n_D + \binom{n_D}{2} \ge 15$. The structure contains no matrices or map from these counts to actual transition edges. In particular, counting pairs of labels does not establish that their relative

transition matrices lie in the Fourier family. The computation $\min(2+1, 3+1) = 3$ is also a numerical evaluation; it supplies no classification of arbitrary complex MU bases.

The existing Route C conclusion has the exact premise

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h_landing : CompleteMUB6 → RouteCCompleteMUBFamily
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The smaller interface FourierTripleLandingPremise removes orbit counts from the contradiction:

$$\forall B : \mathrm{Fin}(7) \to \mathrm{Basis}, \quad \mathrm{PairwiseMU}(B) \Rightarrow$$
$$\exists T : \mathrm{MUTriple}, \quad \mathrm{InConfig}(T, B) \wedge \mathrm{InFourierDitaChart}(T).$$

no_complete_mub6_of_fourier_triple_landing derives the abstract obstruction from this supplied premise. InFourie rDitaChart requires certified domain membership, including analytic soundness. A concrete algebraic landing theorem and a concrete interpretation of the chart remain open.

The proposed expression $\sum_{a,b} |\operatorname{Tr}(B_a^* B_b)|^4$ depends on basis-column phases. By contrast, the entrywise fourth-power sum is fixed by MU: each distinct ordered pair contributes $36(1/6)^2 = 1$, and a self-pair contributes 6. For seven bases its total is $7 \cdot 6 + 7 \cdot 6 = 84$. This computation supplies no Fourier membership map. A separate exact fixture checks that the ordinary raw Fourier matrix has a 2×2 cross ratio ζ_6 with $\zeta_6^4 \neq 1$. Row/column rephasing preserves cross ratios. Thus the proposed all-fourth-root entry condition is not a definition of this Fourier family. Likewise, a simplicial coboundary identity over integer cochains contains no complex transition matrix data unless an additional bridge has been established.

Jaming et al., arXiv:0902.0882 [1] give a Fourier-family quartet obstruction. Applying that literature result to an arbitrary seven-basis family still requires a Fourier-family landing statement. It is not an unconditional classification supplied by the citation.

4. The 56-cell Krawczyk and reflection frontier

canonicalModuliPartitionCover56 stores 56 intervals with endpoints $i/56$ and $(i+1)/56$, a depth label 64, and fixed positive rational margins. These data do not record a depth-64 contractor execution or the three cusp collars.

The kernel-checked equivalence certified_point_in_cover56_iff exposes the actual hypothesis:

$$\mathrm{CertifiedPointInCover56}(\mathrm{cover}, \mathrm{pt})$$
$$\iff \mathrm{GeometricPointInCover56}(\mathrm{cover}, \mathrm{pt})$$
$$\wedge\ \mathrm{HasCurvatureCertificate}(\mathrm{pt.triple}).$$

Thus conjecture3_szollosi_deltoid_closure consumes a curvature certificate; it does not derive that certificate from interval containment.

The native library src/omega_native/mub6_smale_sentry.rs contains rational linear algebra and a Krawczyk contractor for supplied square quadratic systems. The same-named binary under src/bin/ now provides --inspect-system for the exported joint system. This mode performs exact interval inspection, without a depth-64 contractor or a complete cell/cusp run. The terminal receipt has aggregate labels and counts, without 56 per-cell rational boxes, interval Jacobians, or Krawczyk images. Its status label is not an independent replay of those missing objects.

A valid completion requires a concrete chart/system encoding, all cell and collar domains, exact Jacobian bounds, complete coverage, and checked exclusion on each leaf. Krawczyk strict inclusion supports root existence under its hypotheses; exclusion instead requires disjointness or another sound infeasibility condition. A byrfl evaluation of a Boolean establishes that computation. A separate soundness theorem must connect it to the concrete MU residuals and domain.

F_system in the current abstract Lean module repeats a basis-level MU predicate for the row and column indices. It is not yet the 108 scalar coordinate polynomials needed by the proposed native-to-kernel reflection. The cross-basis

residuals also need the candidate's unitarity/domain conditions: a root on an unrestricted column array does not by itself represent a fourth basis.

The source family also needs careful domain identification. Szöllősi, arXiv:0811.3930v2 [5] describes $X_6(\alpha)$ as a two-real-parameter family over an intersection of deltoid regions, containing the one-parameter $B_6(\theta)$ family. The stored scalar interval cells give no coverage map for that full parameter domain. A native certificate must specify the intended family and cover the associated domain.

SzollosiTwoCirculant.lean now checks the continuous block construction $H = \begin{pmatrix} A & B \\ B^* & -A^* \end{pmatrix}$, where the circulant first rows are $(1, x, xy)$ and $(1, u, uv)$. Four unit phases satisfy $\phi(x, y) + \phi(u, v) = 0$, with $\phi(x, y) = x + y + \overline{xy}$. The Gram identity is $HH^* = 6I$, so $H/\sqrt{6}$ is unitary and all its entry moduli squared are one sixth. The parameter domain is exactly six scalar real polynomial equations in eight real coordinates. Its solutions lie in $[-1, 1]^8$, and parameter_covers_intersection connects those solutions to every point of the intersection of the two phase-sum images. The cubic root identities hold without a distinct-root premise. This is the source block construction; equivalence transport to a selected dephased representative and a classification of arbitrary Hadamard matrices are not supplied.

Matolcsi et al., arXiv:2503.14752, Conjecture 3 [4] concerns arbitrary quadruplets containing a member of this family. A single chosen Zauner triplet would not cover all possible third bases. SzollosiQuartetSystem.lean therefore leaves both additional complex bases variable. quartet_iff_candidate_equations retains all 144 of their real coordinates and the actual pairwise MU equations. Every coordinate of a matrix unbiased to the identity lies in the rational box $[-1/2, 1/2]$ by mu_identity_coordinates_bounded. The remaining empty-variety obligation quantifies over every unitary third basis unbiased to both fixed bases. No such variety is established empty. SzollosiQuadraticSystem.lean now gives a joint rational polynomial specification: 440 real variables and 618 equations, each of total degree at most two (equation_degree). Raw matrices are scaled by $\sqrt{6}$ relative to normalized bases; three auxiliary matrices represent their cross products, with entry norm-square target six. The equation counts are:

Equations	Count
Unit phases and complex compatibility	6
Family-matrix ties	72
Raw unitarity of the two additional bases	144
Unit entry norms of the two additional raw bases	72
Three cross-product matrix ties	216
Cross-product entry norms	108

exists_solution_with_parameters_iff checks both directions between actual quartets and solutions at every fixed admissible phase parameter. exists_solution_iff_quartet gives the full existence equivalence. solution_in_full_box bounds every phase and raw basis coordinate in $[-1, 1]$ and every auxiliary cross-product coordinate in $[-3, 3]$. exists_bounded_solution_iff_quartet checks that this rational box loses no solutions. Both additional bases remain arbitrary, including complex phases.

SzollosiExecutableSystem.lean adds executable expression trees for every checked equation. expression_polynomial and equation_eval identify their real evaluation with the rational specification for every assignment. variable_codes and equation_codes check complete ordered IDs for all 440 coordinates and 618 equations. The exporter writes the full rational containing box and three rational probes to mub_szollosi_executable_system.json; its raw SHA-256 is 1367d390da36e5ccf65abbc8fbf529ff07f9e154c277bb14c1cd8c36ba24c947. Independent exports were byte-identical.

The native inspector compiled 5,128 sparse terms of degree at most two, matched all 1,854 probe values, and emitted all 618 full-box residual intervals in mub_szollosi_native_replay_receipt.json. Its observed status is UNRESOLVED_FULL_BOX. Finite probes do not establish native symbolic parity for every real assignment.

SzollosiTermExpansion.lean derives complete sparse normalization semantics. The native inspector now compares all 618 rows and 5,128 nonzero terms against the independently exported Lean coefficient table, raw SHA-256 44656965bb72e735febfc69c1bcd599adbeea2e8512ebd6141025f8524e5d2df. SzollosiNativeReflection.lean checks the actual emitted coefficient table by kernel reduction (native_terms_match); native_coefficients_polynomial identifies every row with the actual rational polynomial under injective coding. SzollosiNativeInterval.lean derives sparse interval

enclosure with square refinement and ordered accumulation. Kernel reduction checks all 618 actual emitted intervals (native_ranges_match); native_interval_enclosed binds each to all actual real assignments in the full containing box. The native receipt raw SHA-256 is 17b31a7eec56d0129b0b4b8a837f55d71b37ee0be0d0a91ad6fbb106949633dc. These identities concern the current emitted data, without generic verification of every Rust compiler or JSON decoder input. Every emitted interval still contains zero (native_full_box_unresolved). No variety emptiness or root existence follows. Coefficient and interval literal mutations are rejected by their kernel checks; a quadratic perturbation invisible to all three old probes is rejected by the complete native coefficient comparison.

SzollosiIntervalExclusion.lean derives rational AST interval enclosure for every contained real assignment. exclusio n_check_sound connects a successful residual interval check to absence of actual polynomial zeros in that box. The full containing-box check reduces to false inside the kernel (full_box_check_unresolved). A concrete first-phase box with both coordinates in $[-1/2, 1/2]$ is excluded by phase_interior_excluded (kernel reduction); this box contains no admissible unit phase and supplies no exclusion of admissible quartets. The complete containing box remains unresolved in both calculations. A false exclusion check supplies no root-existence conclusion.

RectangularQuadraticExclusion.lean now supplies a rational rectangular Krawczyk exclusion method. list_derivativ e_polynomial identifies the computed derivative expressions with formal partial derivatives. The exact quadratic midpoint secant identity yields root_in_image: every zero in the input box lies in its image for any rational preconditioner. No inverse or square projection is required. Strict separation in one coordinate gives exclusion_sound; endpoint contact is retained. A checked three-equation, two-coordinate example separates a coordinate even though every natural residual interval contains zero. A second checked example has strict image inclusion and no common root, so strict inclusion alone supplies no root conclusion for an overdetermined system.

SzollosiRectangularExclusion.lean derives coefficient shape from the checked native table and retains all 618 equations and 440 coordinates. derivative_enclosed encloses formal derivatives of the actual coded quartet polynomials. no_s olution_in_box_of_check applies each successful leaf check to actual polynomial solutions; no_quartet_of_check applies a successful containing- box check to actual quartets. Both require an actual successful rational check and a center inside the box. No successful quartet-domain exclusion or complete 56-cell/cusp run is supplied.

The native rectangular command now emits the full containing-box request, all 618 sparse Jacobian rows (9,000 entries), 440 images and intersections. Its deterministic preconditioner has 28 nonzero rational entries. The zero center lies in the containing box and need not satisfy the phase equations. The output in artifacts/mub_szollosi_rectangular_nativ e_receipt.json (raw SHA-256 eccd3b1c075d0058cf729e2aa0af5e210860da93d2178ea09a9d5cfb3b820a76) is CONT RACTED_UNRESOLVED_BOX: 24 coordinates contract and none separates. These contractions restore constant entries of the constructed family matrix; none of the 144 raw coordinates of the two additional bases is narrowed by this run, as checked by native_additional_bases_unchanged. The saved request reproduces the receipt byte for byte.

SparseRectangularExclusion.lean derives finite-support image soundness. SzollosiRectangularReflection.lean identifies every emitted derivative interval and image with the actual polynomial definitions, using bounded row checks evaluated inside the Lean kernel. native_jacobian_entry_enclosed bounds actual formal derivatives. solution_in_nati ve_retained retains every actual polynomial solution in every recorded intersection. The input box, zero center, center residuals, contracted-coordinate list and non-separation are also kernel checked. These checks apply to the current emitted data; generic Rust/JSON verification is not supplied. No root existence follows. SzollosiColumnGauge.lean now derives an independent column-phase reduction for both additional bases. It leaves the constructed family matrix and all eight phase coordinates fixed, and preserves the raw unitary, unit-entry and all three MU Gram-norm equations. Conjugating each additional basis's first-row entries sets that row to one. The same 618-equation system therefore has a representative with 24 prescribed real coordinates whenever a quartet exists, also at every fixed admissible family parameter. These coordinates are wire IDs 80-91 and 152-163, with alternating real value one and imaginary value zero; their complete ordered list is checked by gaugePointCodes_checked using byrfl. gauge_box_firstRow derives the first-row equalities from point-box membership; exists_gauge_box_solution_with_parameters_iff gives the exact existence reduction with no separate gauge premise. The bounds are exported in artifacts/mub_szollosi_column_gauge_box.json. This normalizes columns of the two variable bases; it does not classify or fully dephase the family matrix. The earlier full-box contractor receipt is unchanged. No quartet-domain exclusion or complete cell/cusp coverage follows from this reduction.

SzollosiGaugeBalance.lean derives row and column sums of six for the raw transition between the two additional bases after the column gauge. SzollosiGaugedExecutable.lean retains all 618 original equations and appends 24 gauge

equations and 24 derived balance equations. The resulting 666 quadratics use the same 440 coordinates and preserve quartet existence at every fixed admissible family parameter. The balance equations are conclusions of the original equations and gauge, rather than new assumptions on quartets. The Lean export and native replay agree on all 666 coefficient rows, 5,320 sparse terms and 1,998 rational probe values; the first 618 rows are unchanged.

SzollosiGaugedCover.lean checks a six-chart solution cover: at least one real entry in the first transition row is at least one, since that row sums to six. The six charts overlap. SzollosiGaugedReflection.lean connects the actual native chart bounds to every augmented-system solution, also at each admissible family parameter, using the earlier checked retained-box theorem. Both additional bases remain variable. Duplicate native runs on all six charts return UNRESOLVED_BOX; none of these charts has been excluded.

A separate candidate box restricts all six real entries of that transition row to [-1/2,1/2]. Its balance equation forces the first real entry into [7/2,17/2], separated from its own input interval. The native receipt artifacts/mub_szollosi_balance_exclusion_receipt.json records this image at coordinate 368. Kernel checks bind the 48 added coefficient rows and all 440 input bounds; the selected image has independent real semantics from the balance identity. A contained point has admissible phases (1,-1,1,i), so this box's exclusion is not an off-phase-domain example. That point is not a quartet. The result excludes this candidate box, without excluding all candidates at any family parameter. Other newly emitted images and the generic Rust/JSON runtime are not verified by this selected-image result. The new root replay fixes 48 coordinates (24 constructed-family constants and the 24 column gauges) with 74 nonzero preconditioner entries. It remains unresolved. Six candidate charts do not replace the required 56 family cells and three cusp collars, and HasCurvatureCertificate remains an explicit premise.

Useful preconditioners, further subdivision, further equivalence transports, complete parameter/candidate coverage, cusp analysis and curvature remain open. The native runtime receipt retains its kernel_reflection_complete:false field; the separate checkpoint binds the subsequent Lean checks without altering the native output.

The Lean SmalePointCertificate and KrawczykContractor structures already contain a root, its equation, and uniqueness evidence. The current existence lemmas project these fields; krawczyk_inclusion_criterion does not derive the root from K_subset_interior. They therefore supply no independent analytic soundness bridge for a native contractor receipt.

The separate deltoid-cover implementation restricts candidates to canonical_candidate_basis_box(), whose first real component lies in $[9/10, 1]$. Consequently $|U_{0,0}|^2 \geq 81/100$, and its base margin is $(81/100 - 1/6)^2 = 37249/90000$. Every basis unbiased to the standard basis has $|U_{0,0}|^2 = 1/6$, outside that box. This margin therefore excludes a restricted candidate domain that contains no such unbiased basis. Subdividing the family parameter domain, even exhaustively, does not enlarge the candidate domain. The restriction is explicit in mub6_szollosi_deltoid_cover.rs::canonical_candidate_basis_box and SzollosiDeltoidCoverReport.candidate_domain_restriction.

5. The flat-extension and operator frontier

For a genuine normalized moment functional L, a degree-two moment matrix has entries $M_2(L)_{u,v} = L(u^*v)$ for a declared word basis. A flat-extension bridge must bind M_1 to the corresponding principal submatrix, require positivity and relation compatibility, and establish actual equality of ranks. The legacy rank labels in RankLoopClosure and CurtoFialkowFlatExtension are not computed ranks of these matrices. The added rationalMomentMatrix evaluates explicit word products in $\mathbb{Q}$. MomentPrincipalEmbedding binds the lower word basis to a principal submatrix, and rationalMomentMatrix_rank_mono establishes rank monotonicity. FlatMomentRankCondition now states equality of the actual degree-two and degree-one matrix ranks, together with this inclusion and complete, nonduplicate indexing of all words at both degrees. Empty index lists cannot satisfy it. No MUB-specific flatness witness, relation compatibility, or extraction follows from that definition.

gns_extraction_theorem takes a structure already carrying the operators and state. It repackages that data into the present realization interface. The legacy theorem supplies no operator extraction from moment values. The rank-loop exclusion theorem retains h_gns_exhaust; the emptiness variants retain their soundness or trace-sign arguments.

There is a preceding domain defect. PhysicalProjectorRealization does not require basis completeness, rank-one ranges, Hilbert dimension six, inner-product linearity, or cross-basis unbiasedness. zeroActionInterfaceModel supplies

a Boolean carrier, zero actions, and a normalized designated state satisfying every current field. The theorem not_universal_interface_exclusion then gives

```
¬ UniversalPhysicalProjectorExclusion numBases dim
```

for arbitrary size labels, with no axiom dependencies. This is an interface countermodel, not a concrete quantum or MUB realization. It prevents treating the present exclusion predicate as the requested physical theorem.

ConcreteMubProjectors.lean provides a separate forward construction from FiniteComplexMubFamilyιBι. Its actual complex matrices are $P_{b,l} = v_{b,l} v_{b,l}^*$. The checked lemmas establish self-adjointness, same-basis multiplication, $\sum_l P_{b,l} = I$, trace one, rank one, and $\mathrm{Tr}(P_{b,l} P_{c,k}) = 1/|\iota|$ for distinct bases. This constructs the projectors from a supplied complex family; it neither supplies such a family in dimension six nor extracts it from positive truncated moments. The legacy interface countermodel therefore remains relevant.

ConcreteMubMoments.lean now constructs complex word moments directly from that supplied family. Choose a state $\psi = v_{b_0,l_0}$ and a distinct basis $b_1 \neq b_0$. For a word w, let P_w be the ordered projector product and $L_\psi(w) = \langle \psi, P_w \psi \rangle$. The complete, nonduplicate word lists of lengths at most one and two index actual Hermitian matrices. The theorem complexMomentMatrix_eq_gram identifies each matrix with the Gram matrix of its word-state columns; complexMomentMatrix_posSemidef establishes positivity. complexMomentMatrix_principal_degree_one identifies the lower principal submatrix. MU makes every overlap with basis b_1 nonzero, so its singleton projector columns span the coordinate space. Consequently concrete_mub_moment_flatness gives $\mathrm{rank}(M_2) = \mathrm{rank}(M_1) = |\iota|$. Normalization at the empty word is also checked.

This is a forward theorem for moments induced by a supplied concrete family, with the distinct-basis and state labels explicit. It supplies neither a family in dimension six nor a reverse extraction from arbitrary positive flat data. The rational moment API remains separate; an adequate MUB interpretation of abstract positive flat data remains required.

ConcreteMubPolynomials.lean now supplies actual coordinate polynomials for a fixed triple of complex matrices. A candidate $U \in \mathbb{C}^{6\times 6}$ is encoded by 72 real coordinates, with a checked inverse decoding. The 108 MU residuals are $r_{a,i,j} = |\langle T_{a,i}, U_j \rangle|^2 - 1/6$. The 72 real/imaginary residuals of $U^*U - I$ impose candidate orthonormality. These are 180 indexed constraints, without an independence claim. polynomialIdeal is their generated ideal in MvPolynomialCoord over $\mathbb{R}$; polynomialIdeal_le_eval_kernel_iff identifies its evaluation-kernel condition with their simultaneous vanishing. muPolynomial_eval_projector_trace identifies each MU residual with the corresponding complex projector cross-trace minus one sixth.

polynomialVariety_eq_empty_iff connects this actual real zero set to absence of a unitary candidate with the stated MU overlaps. A supplied complete complex MUB family gives a point in the zero set of its first three bases, using its fourth basis as candidate, through family_triple_variety_nonempty. This is a checked semantic bridge. It does not establish zero-set emptiness. The separate commutative construction below now extracts a candidate from normalized positive flat coordinate moments satisfying the finite residual equations. The fixed triple's coefficients are real and imaginary parts of complex entries; rational enclosures over the continuous family cells and cusp collars still require separate construction.

FlatMomentProjectorExtraction.lean now gives a finite reverse construction. Its inputs are normalized complex word values, positivity of the actual complete degree-two moment matrix, equality with the actual degree-one matrix rank, and contextual idempotence equations $L(u^* g g v) = L(u^* g v)$ for each generator and lower-degree words. It constructs a Gram factor, one common complex linear space, a normalized state, and linear idempotent actions symmetric for the explicit finite Hermitian pairing. extract_word_projectors identifies every degree-two moment entry with $L(u^* v) = \langle P_u \psi, P_v \psi \rangle$, for words of length at most two. The space dimension is $\mathrm{rank}(M_1)$. Operators and a state are outputs, rather than input structure fields.

concrete_moment_extraction_inputs checks that the previously constructed moments from a supplied MUB family satisfy these extraction premises. FlatMomentBasisRelations.lean adds the exact contextual equations $L(u^* g_{b,l} g_{b,k} v) = \delta_{lk} L(u^* g_{b,l} v)$ and $\sum_l L(u^* g_{b,l} v) = L(u^* v)$ for every basis and all words of length at most one on either side. These equations are input relations, rather than consequences of positivity alone.

From these relations, extract_word_basis constructs actions satisfying $P_{b,l} P_{b,k} = \delta_{lk} P_{b,l}$ and $\sum_l P_{b,l} = I$ on the common extracted space, while retaining symmetry, state normalization, and every degree-two moment entry. concrete_moment_basis_inputs checks the contextual relations for moments of a supplied MUB family. The normalized

word assignment that is zero on every nonempty word fails completeness, as checked by zero_action_moments_reject_completeness.

For arbitrary input word data, dimension six, nonzero rank-one ranges for every outcome, and cross-basis MU conclusions still need their corresponding relations and derivations. The finite representation does not assert a full star-algebra extension or transport arbitrary word data to the coordinate polynomial zero set. These remain limits of that noncommutative word route. The real coordinate route below supplies a separate extraction and its empty-variety consequence without that additional word-to-coordinate transport.

CommutativeFlatMoments.lean constructs real coordinate multiplication operators directly. Let the lower index list contain the constant and every coordinate, and the upper list also contain every ordered coordinate product. The repeated quadratic monomials are retained. For a real linear functional Λ, the actual entries are $M_2(\Lambda)_{i,j} = \Lambda(m_i m_j)$. Normalization $\Lambda(1) = 1$, positive semidefiniteness and the actual equality rank $M_2 =$ rank M_1 are explicit premises.

A Gram factor and rank equality identify the full column space with its lower column space. The coordinate shifts descend to symmetric endomorphisms there; polynomial commutativity yields commuting operators. Their joint real eigenspaces span this finite space. A joint eigenvector with nonzero overlap with the normalized state yields one common real point. Every quadratic polynomial p satisfying $\Lambda(p^2) = 0$ vanishes at that point, through real_point_of_positive_flat. The operators and point are constructed, rather than supplied as extraction fields.

ConcreteMubFlatExtension.lean checks that all 180 candidate-basis residuals r_e lie in this quadratic span. FiniteRelationCompatibility states $\Lambda(m_i r_e) = 0$ for every upper-list monomial and every residual. These finite equations imply $\Lambda(r_e^2) = 0$, so the same extracted point annihilates all residuals. fourth_basis_of_positive_flat produces an actual complex unitary candidate with all 108 MU overlaps for the fixed triple.

Evaluation at any actual solution supplies a normalized positive moment matrix with both ranks equal to one and satisfies every finite relation equation. fourth_basis_iff_positive_flat_moments gives the resulting two-way equivalence between candidate existence and existence of normalized positive flat compatible coordinate moments. moment_nonpositive_of_empty_variety consequently excludes positive semidefiniteness when the fixed real variety is empty, retaining normalization, actual flatness and finite relation compatibility as hypotheses. It does not exclude arbitrary nonflat truncations or establish variety emptiness.

The concrete flat-moment, normalized-quartet, and joint polynomial equivalences were freshly replayed as dependencies of the 2026-10-11 transport bridge, with only propext, Classical.choice, and Quot.sound in their reported axioms. This scoped replay covers 23 local source modules and cached dependencies; it is not a fresh build of the complete MUB6 import surface. This real coordinate model is separate from the rational and noncommutative word APIs; no implicit identification of their data is used.

Laurent and Mourrain, arXiv:0812.2563 [6] and Mourrain and Schmüdgen, arXiv:1406.4975 [7] provide commutative and star-algebra flat-extension settings, respectively. The checked finite commutative construction above derives its own operator and point extraction from explicit hypotheses. A full noncommutative extension is not inferred from those references. Variety emptiness does not, by itself, exclude an arbitrary fixed truncation; normalization, actual flatness and relation compatibility remain part of the stated consequence.

6. Receipt custody and validation

The R6 receipt is present and has raw SHA-256 cf8442d8fe1397226d669fcea7bc038ef8d984a9bae741772b9d742fbe62eb59. It records NOT_A_CERTIFICATE.

The Matolcsi receipt has raw SHA-256 598d73b4f7b6807d62fc8d5c7698fd7c00ca73634c24641e0f2a848c873fc6fa; its internal digest is a different commitment. The historical zenodo/release_manifest.sha256 contains 496 entries: 493 archive file members plus three release-metadata files. The archive extraction regression checks all 493 file members. Digest agreement establishes the corresponding byte or normalization commitment, rather than theorem validity.

The research source snapshot is pinned at commit 2d03b504106356593d76f5a6bafc259019d52247. The Lean source package and Rust workspace are external research materials. The preprint ZIP does not bundle those trees, their dependencies or a new axiom census. Its integrity check verifies only the delivered manuscript package. The historical research archive and receipt counts above describe a separate artifact, not the preprint ZIP.

Mub6 is the canonical package root; the external-pinned integration target is separate. Mub6 currently imports MubBal ancedTable, whose full build has not completed (Section 1.4), so a successful lakebuildMub6 has not been recorded for this checkpoint. The full build and all-declaration axiom audit must complete before a package-wide formal validation claim is made. The live execution evidence and outstanding gates are recorded in the executive checkpoint report; historical test totals are not substituted for current runs. The 2026-10-11 finite-corner receipt separately binds the canonical index/minor targets, the external public theorem, the 22-source integration replay, and their printed axiom reports. It does not assert a whole-package replay. The subsequent bounded-quartet receipt binds the two canonical block-chart and division-free equation targets, their 34 endpoint reports, and the rejected determinant-conjugation mutation.

The bounded export receipt binds 31 local source modules, 28 displayed index, collection and coded-system endpoints, two rejected mutations, and the complete native coefficient artifact. Its source-level equivalence is kernel-checked; serialized-byte reflection and global infeasibility remain open. These scoped receipts do not report execution of the complete validation command block above.

7. Publication checkpoint and completion requirements

The package is not ready for an arXiv submission asserting mathematical closure, or for a new Zenodo release with that claim. A scoped technical checkpoint must still complete rendering, metadata reconciliation, semantic review, and archive re-certification before deposit. The existing archive is historical and does not contain the current edits by implication.

The full Option B objective is retained: establish a concrete Route C landing bridge or state its defensible remaining premises; execute and soundly reflect the complete 56-cell/cusp exclusion calculation; construct the operator moment and flat-extension bridge in an adequate mathematical domain. A passing suite or a standard-axiom implication cannot replace any of those requirements.

8. Mechanism inventory and evidence boundaries

Mechanism	Mathematical object or conclusion	Evidence boundary
Concrete normalization and transport	Raw and normalized quartet equivalence	Reported external Lean endpoints; a current full replay remains required
Quantitative minor and block chart	Squared determinant at least 54/5; inverse-entry squared moduli at most 10/27	Section 1.3 and named external Lean endpoints; no nonextension conclusion
Balanced polynomial reduction	648 equations in 320 bounded real coordinates	Reported degree and equivalence endpoints; infeasibility open
Fourier-family route	Conditional contradiction from a Fourier-triple landing premise	The landing and concrete chart-soundness premises remain open
Interval exclusion and six charts	Sound exclusion criteria and recorded candidate boxes	No complete admissible-family exclusion or 56-cell/cusp replay
Positive flat moment extraction	Fourth-basis construction from explicit compatible moment hypotheses	Empty varieties and flatness for proposed data remain open
Finite Tao calculation	Norm 9 on all twenty balanced sign patterns	Finite matrix evaluation; no general Laurent necessary-condition theorem
Authorization	None	Integrity checks and reported computations confer no mathematical or publication approval

The retained research ledgers are historical working indexes in the source snapshot. They are not part of the standalone

preprint or its verification contract. The source theorem names in Sections 1–5 specify the individual reported results; their stated hypotheses remain part of each statement.

References

1. P. Jaming, M. Matolcsi, P. Móra, F. Szöllősi and M. Weiner, "A generalized Pauli problem and an infinite family of MUB-triplets in dimension 6", arXiv:0902.0882.
2. A. Maxwell and S. Brierley, "On properties of Karlsson Hadamards and sets of Mutually Unbiased Bases in dimension six", arXiv:1402.4070.
3. M. Cárdenes Wuttig and J. Tindall, "A Complete Classification of Complex Hadamard Matrices of Order Six", arXiv:2608.18053v2.
4. M. Matolcsi, Á. K. Matszangosz, D. Varga and M. Weiner, "Triplets of mutually unbiased bases", Journal of Algebraic Combinatorics (2026), doi:10.1007/s10801-026-01506-x; arXiv:2503.14752. Cited in Sections 2.1 and 4 under both identifiers.
5. F. Szöllősi, "A two-parameter family of complex Hadamard matrices of order 6 induced by hypocycloids", arXiv:0811.3930v2.
6. M. Laurent and B. Mourrain, "A Sparse Flat Extension Theorem for Moment Matrices", arXiv:0812.2563.
7. B. Mourrain and K. Schmüdgen, "Flat extensions in *-algebras", arXiv:1406.4975.